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Introduction

These sample exam questions were originally included in the *AP Calculus AB and AP Calculus BC Curriculum Framework,* published in fall 2014. The *AP Calculus AB and AP Calculus BC Course and Exam Description,* which is out now, includes that curriculum framework, along with a new, unique set of exam questions. Because we want teachers to have access to all available questions that support the new exam, we are making those from the fall 2014 curriculum framework available in this supplementary document.

The sample exam questions illustrate the relationship between the curriculum framework and the redesigned *AP Calculus AB Exam and AP Calculus BC Exam*, and they serve as examples of the types of questions that appear on the exam.

Each question is accompanied by a table containing the main learning objective(s), essential knowledge statement(s), and Mathematical Practices for AP Calculus that the question addresses. For multiple-choice questions, an answer key is provided. In addition, each free-response question is accompanied by an explanation of how the relevant Mathematical Practices for AP Calculus can be applied in answering the question. The information accompanying each question is intended to aid in identifying the focus of the question, with the underlying assumption that learning objectives, essential knowledge statements, and MPACs other than those listed may also partially apply. Note that in the cases where multiple learning objectives, essential knowledge statements, or MPACs are provided for a multiple-choice question, the primary one is listed first.

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AP Calculus AB Sample Exam Questions

Multiple Choice: Section I, Part A

A calculator may not be used on questions on this part of the exam.

1. $\lim\limits_{x \to \pi} \dfrac{\cos x + \sin(2x) + 1}{x^2 - \pi^2}$ is

(A) $\dfrac{1}{2\pi}$

(B) $\dfrac{1}{\pi}$

(C) 1

(D) nonexistent

Learning Objectives	Essential Knowledge	Mathematical Practices for AP Calculus
LO 1.1C: Determine limits of functions.	**EK 1.1C3:** Limits of the indeterminate forms $\frac{0}{0}$ and $\frac{\infty}{\infty}$ may be evaluated using L'Hospital's Rule.	**MPAC 1:** Reasoning with definitions and theorems **MPAC 3:** Implementing algebraic/computational processes
LO 2.1C: Calculate derivatives.	**EK 2.1C2:** Specific rules can be used to calculate derivatives for classes of functions, including polynomial, rational, power, exponential, logarithmic, trigonometric, and inverse trigonometric.	

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2. $\lim_{x \to \infty} \frac{\sqrt{9x^4 + 1}}{x^2 - 3x + 5}$ is

(A) 1

(B) 3

(C) 9

(D) nonexistent

Learning Objectives	Essential Knowledge	Mathematical Practices for AP Calculus
LO 1.1C: Determine limits of functions.	**EK 1.1C2:** The limit of a function may be found by using algebraic manipulation, alternate forms of trigonometric functions, or the squeeze theorem.	**MPAC 3:** Implementing algebraic/computational processes **MPAC 2:** Connecting concepts
LO 1.1A(b): Interpret limits expressed symbolically.	**EK 1.1A2:** The concept of a limit can be extended to include one-sided limits, limits at infinity, and infinite limits.	

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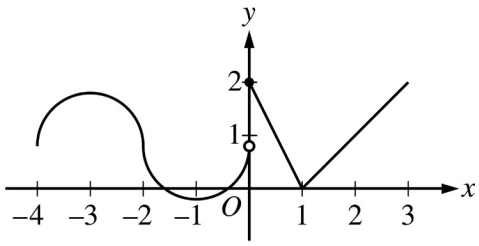

Graph of f

3. The graph of the piecewise-defined function f is shown in the figure above. The graph has a vertical tangent line at $x = -2$ and horizontal tangent lines at $x = -3$ and $x = -1$. What are all values of x, $-4 < x < 3$, at which f is continuous but not differentiable?

(A) $x = 1$

(B) $x = -2$ and $x = 0$

(C) $x = -2$ and $x = 1$

(D) $x = 0$ and $x = 1$

Learning Objectives	Essential Knowledge	Mathematical Practices for AP Calculus
LO 2.2B: Recognize the connection between differentiability and continuity.	**EK 2.2B1:** A continuous function may fail to be differentiable at a point in its domain.	**MPAC 4:** Connecting multiple representations **MPAC 2:** Connecting concepts
LO 1.2A: Analyze functions for intervals of continuity or points of discontinuity.	**EK 1.2A3:** Types of discontinuities include removable discontinuities, jump discontinuities, and discontinuities due to vertical asymptotes.	

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4. An ice sculpture in the form of a sphere melts in such a way that it maintains its spherical shape. The volume of the sphere is decreasing at a constant rate of 2π cubic meters per hour. At what rate, in square meters per hour, is the surface area of the sphere decreasing at the moment when the radius is 5 meters? (Note: For a sphere of radius r, the surface area is $4\pi r^2$ and the volume is $\frac{4}{3}\pi r^3$.)

(A) $\frac{4\pi}{5}$

(B) 40π

(C) $80\pi^2$

(D) 100π

Learning Objectives	Essential Knowledge	Mathematical Practices for AP Calculus
LO 2.3C: Solve problems involving related rates, optimization, rectilinear motion, *(BC)* and planar motion.	**EK 2.3C2:** The derivative can be used to solve related rates problems, that is, finding a rate at which one quantity is changing by relating it to other quantities whose rates of change are known.	**MPAC 2:** Connecting concepts **MPAC 3:** Implementing algebraic/computational processes
LO 2.1C: Calculate derivatives.	**EK 2.1C5:** The chain rule is the basis for implicit differentiation.	

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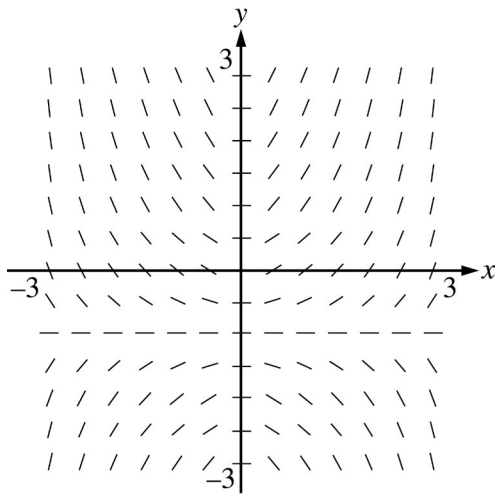

5. Shown above is a slope field for which of the following differential equations?

(A) $\frac{dy}{dx} = xy + x$

(B) $\frac{dy}{dx} = xy + y$

(C) $\frac{dy}{dx} = y + 1$

(D) $\frac{dy}{dx} = (x+1)^2$

Learning Objective	Essential Knowledge	Mathematical Practices for AP Calculus
LO 2.3F: Estimate solutions to differential equations.	**EK 2.3F1**: Slope fields provide visual clues to the behavior of solutions to first order differential equations.	**MPAC 4**: Connecting multiple representations **MPAC 2**: Connecting concepts

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$$f(x)=\begin{cases}2x-2 & \text{for } x<3\\ 2x-4 & \text{for } x\geq 3\end{cases}$$

6. Let f be the piecewise-linear function defined above. Which of the following statements are true?

I. $\lim\limits_{h\to 0^-}\dfrac{f(3+h)-f(3)}{h}=2$

II. $\lim\limits_{h\to 0^+}\dfrac{f(3+h)-f(3)}{h}=2$

III. $f'(3)=2$

(A) None

(B) II only

(C) I and II only

(D) I, II, and III

Learning Objectives	Essential Knowledge	Mathematical Practices for AP Calculus
LO 2.1A: Identify the derivative of a function as the limit of a difference quotient.	**EK 2.1A2:** The instantaneous rate of change of a function at a point can be expressed by $\lim\limits_{h\to 0}\dfrac{f(a+h)-f(a)}{h}$ or $\lim\limits_{x\to a}\dfrac{f(x)-f(a)}{x-a}$, provided that the limit exists. These are common forms of the definition of the derivative and are denoted $f'(a)$.	**MPAC 2:** Connecting concepts **MPAC 5:** Building notational fluency
LO 1.1A(b): Interpret limits expressed symbolically.	**EK 1.1A2:** The concept of a limit can be extended to include one-sided limits, limits at infinity, and infinite limits.	

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